Chapter

Mechanics

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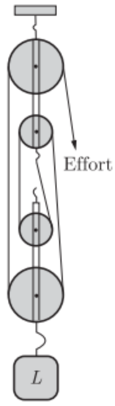
1. The given diagram shows the movable block of a block and tackle system with effort in a convenient direction. From the diagram we can conclude that the number of pulleys used in the fixed block are ____________.



(a) 1

(b) 3

(c) 2

(d) 4

SQP 2025

2. A ___________ (class I/class II/Class III) lever will always have M.A. > 1.

SQP 2025

3. In a block and tackle system, increase in the weight of the movable block ___________ (decreases, does not affect, increases) the efficiency of the pulley system.

SQP 2025

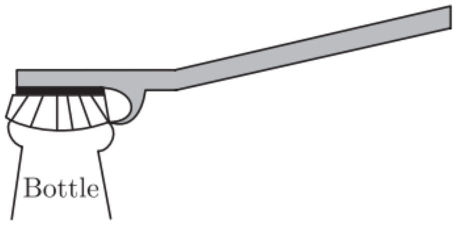
4. A block and tackle system of pulleys has velocity ratio 4.

(a) Draw a labelled diagram of the system indicating clearly, the direction of the load and effort.

(b) Calculate the potential energy of the load 100 kgf lifted by this pulley to a height 5 m. ($g = 10\ ms^{-2}$)

SQP 2025

5. Name the class of the lever shown in the picture below:



MAIN 2024

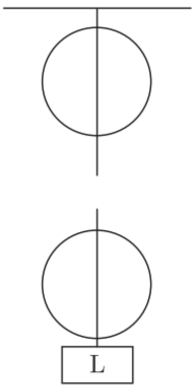
6. Meera choose to use a block and tackle system of '9' pulleys instead of a single movable pulley to lift a heavy load.

(a) What is the advantage of using a block and tackle system over a single movable pulley?

(b) Why should she connect more number of pulleys in the upper fixed block?

MAIN 2024

7. Two pulleys are given as below.



(a) Copy and complete the labelled diagram connecting the two pulleys with a tackle to obtain Velocity Ratio = 2.

(b) If Load = 48 kgf and efficiency is 80% then calculate:

1. Mechanical Advantage.

2. Effort needed to lift the load.

MAIN 2024

8. A single movable pulley has :

(a) velocity ratio 2 and actual mechanical advantage 2

(b) velocity ratio 2 and actual mechanical advantage more than 2

(c) velocity ratio 2 and actual mechanical advantage less than 2

(d) none of the above

SQP 2023

9. A labourer uses a sloping wooden plank of length 2.0 in to push up a load of 600 N into the truck at a height of 0.8 m.

(i) What is the mechanical advantage of the sloping plank ?

(ii) How much effort is needed to push the load up into the truck ?

(iii) What assumption have you made in arriving at your answer in part (ii) above?

MAIN 2023

10. What do you understand by the term load? **COMP 2023**

11. The MA of a machine is 5 and its efficiency is 80%. It is used to lift a load of 200 kgf to a height of 20 m. What is the effort required ?

(a) 20 kgf (b) 10 kgf

(c) 50 kgf (d) 40 kgf

SQP 2022

12. Increase in velocity ratio and mechanical advantage being constant would .... the efficiency.

(a) increase (b) decrease

(c) not affect (d) none of these.

SQP 2022

13. Explain why scissors for cutting cloth may have blades much longer than the handles; but shears for cutting metals have short blades and long handles.

SQP 2022

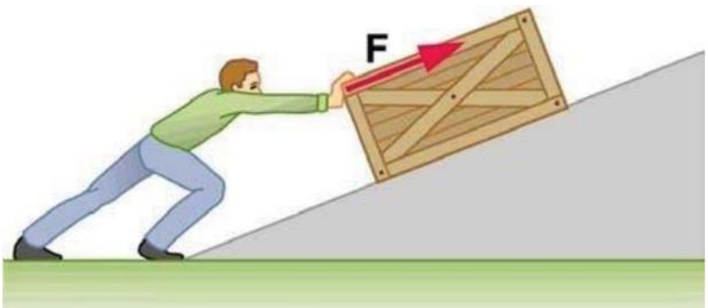
14. Differentiate between ideal mechanical advantage and actual mechanical advantage.

MAIN 2022

15. **With reference to the terms mechanical advantage, velocity ratio and efficiency of a machine, name and define the term that will not change for a machine of a given design.**

COMP 2022

16. **A coolie is pushing a box weighing 1500 N up an inclined plane 7.5 in long on to a platform, 2.5 m above the ground.**



(i) Calculate the mechanical advantage of the inclined plane.

(ii) Calculate the effort applied by the coolie.

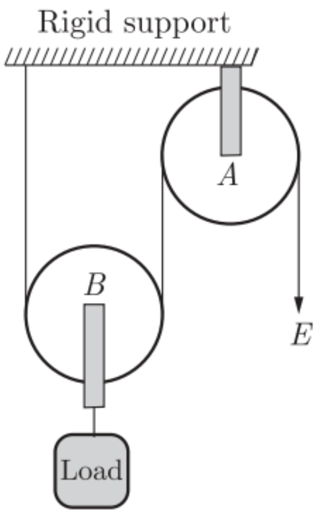
(iii) In actual practice, the coolie needs to apply more effort than what is calculated. Give one reason why you think the coolie needs to apply more effort.

COMP 2022

17. **Give two reasons why the efficiency of a single movable pulley system is not 100%.**

MAIN 2021

18. **In the diagram shown below, the velocity ratio of the arrangement is**



(a) 1 (b) 2

(c) 3 (d) 0

SEM-I, 2021

19. A single fixed pulley is used because

(a) it changes the direction of applied effort conveniently

(b) it multiplies speed

(c) it multiplies effort

(d) its efficiency is 100%

SEM-I, 2021

20. A woman draws water from a well using a fixed pully. The mass of the bucket and the water together is 10 kg. The force applied by the woman is 200 N. The mechanical advantage is (Take $g = 10\ m/s^2$)

(a) 2 (b) 20

(c) 0.05 (d) 0.5

SEM-I, 2021

21. A single fixed pulley is a modified form of ....

(a) Class III lever (b) Frisbee

(c) Class I lever (d) Class II lever

COMP 2021

22. Calculate the power of a pump which can lift 400 kg of water to store it in a water tank at a height of 19 m in 40 s. (Take $g = 10\ ms^{-2}$)

COMP 2021

Solution

1.

Ans: (c) 2

Explanation:

In the diagram, you see three pulleys in the movable block (with the load), and the rope is arranged so that the effort is applied in a convenient (downward) direction. For such a block and tackle, the number of pulleys in the fixed block must match to complete the system; here it comes out to **2 pulleys in the fixed block**.

2.

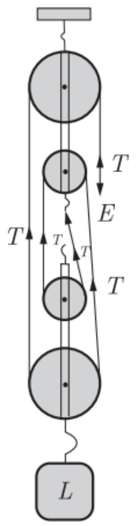
Ans: Class II

3.

Ans: Decreases

4.

Ans:



(a) A correct labelled diagram of a block and tackle system with velocity ratio 4 should show:

- Two fixed pulleys and two movable pulleys,
- The load (L) attached to the lower pulley block,

- The effort (E) applied downward/upward as shown,
- Tension (T) in each segment of the string.

(b) Potential energy of the load:

$$U = mgh = 100 \times 10 \times 5 = 5000 \text{ J}$$

So, the potential energy gained is **5000 J**.

5.

Ans: Class II lever.

6.

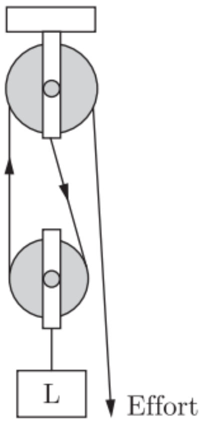
Ans:

(a) A block and tackle system of 9 pulleys gives a **mechanical advantage of 9**, which is much greater than the mechanical advantage of **2** provided by a single movable pulley. Therefore, using a block and tackle makes lifting a heavy load much easier.

(b) Adding more pulleys in the fixed block **increases the mechanical advantage**, which reduces the effort needed to lift the load.

7.

Ans: (a)



(b)

Given:

Load = 48 kgf

Efficiency = 80% = 0.8

Velocity Ratio (V.R.) = 2

1. **Mechanical Advantage (M.A.)**

$$\text{Efficiency} = \frac{\text{M.A.}}{\text{V.R.}}$$

$$0.8 = \frac{\text{M.A.}}{2}$$

$$\text{M.A.} = 1.6$$

2. **Effort needed to lift the load**

$$\text{Effort} = \frac{\text{Load}}{\text{M.A.}} = \frac{48}{1.6} = 30 \text{ kgf}$$

Effort required = 30 kgf

8.

Ans: (c) velocity ratio 2 and actual mechanical advantage less than 2

Explanation:

For a single movable pulley (ideal case), **velocity ratio (VR) = 2** and **mechanical advantage (MA) = 2**.

But in real pulleys, friction and weight of the pulley reduce the useful effort, so the **actual mechanical advantage is slightly less than 2**.

9.

Ans: (i) Mechanical Advantage,

$$\text{M.A.} = \frac{\text{length}}{\text{height}} = \frac{2}{0.8} = 2.5$$

(ii) Effort required,

$$\text{Effort} = \frac{\text{Load}}{\text{M.A.}} = \frac{600}{2.5} = 240 \text{ N}$$

(iii) The assumption made is that there is **no friction** between the plank and the load.

10.

Ans: The load is the resistance or opposing force (L) that the machine has to overcome.

11.

Ans: (d) 40 kgf

Explanation:

Mechanical advantage:

$$\mathrm{MA} = \frac{\text{Load}}{\text{Effort}} = 5$$

Load = 200 kgf

$$5 = \frac{200}{E} \Rightarrow E = \frac{200}{5} = 40 \text{ kgf}$$

So, the effort required is **40 kgf**.

12.

Ans: (b) decrease

$$\text{Efficiency } \eta = \frac{\text{Mechanical advantage}}{\text{Velocity ratio}}$$

If **MA is constant** and **VR increases**, the denominator becomes larger, so the value of η becomes **smaller**.

Hence, efficiency **decreases**.

13.

Ans: Scissors for cutting cloth and shears for cutting metals are both **class I levers**. In class I levers, the mechanical advantage can be **greater than, less than, or equal to 1**.

Cutting cloth does **not require much force**, so the scissors can have **long blades and short handles**. But cutting metal requires **much more force**, so shears are designed with **short blades and long handles** to get a **greater mechanical advantage**.

Hence, scissors for cloth have long blades, while metal-cutting shears have long handles.

14.

Ans:

Ideal Mechanical Advantage (IMA)	**Actual Mechanical Advantage (AMA)**
It is the ratio of the **total load moved** to the **effort applied**, assuming there is **no friction** and the machine is perfect.	It is the ratio of the **useful load lifted** to the **actual effort applied**, considering all real conditions like friction.
$\mathrm{IMA} = \frac{L}{E}$	$\mathrm{AMA} = \frac{l}{E}$

Ideal MA is always **greater than or equal to** actual MA because friction reduces the effectiveness of real machines.

15.

Ans: The term that does **not change** for a machine of a given design is the **Velocity Ratio (V.R.)**.

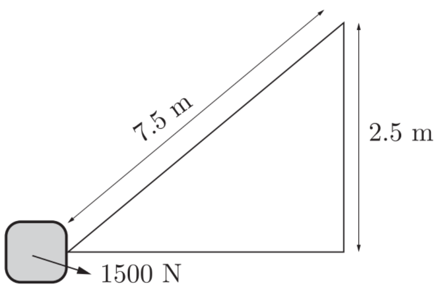
Definition:

Velocity ratio is the ratio of the **distance (or velocity) moved by the effort** to the **distance (or velocity) moved by the load**.

It remains constant because it depends only on the design and geometry of the machine.

16.

Ans:



(i) Mechanical Advantage (M.A.)

$$\text{M.A.} = \frac{\text{Length of the plane}}{\text{Height}} = \frac{7.5}{2.5} = 3$$

(ii) Effort applied:

$$\text{Effort} = \frac{\text{Load}}{\text{M.A.}} = \frac{1500}{3} = 500 \text{ N}$$

(iii) The coolie needs to apply more effort in practice because friction acts on the inclined plane, reducing efficiency. Hence the actual effort becomes more than the calculated (ideal) effort.

17.

Ans: The efficiency of a single movable pulley system is not 100% because:

(i) **The movable pulley has weight**, and this extra weight increases the effort needed.

(ii) **There is friction between the pulley and the string**, which reduces the output.

18.

Ans: (b) 2

Explanation:

In this system there is **one movable pulley** (B). The load is supported by **two segments of the same rope**.

So, when the effort end of the rope moves down by distance $2d$, the load rises by distance d.

$$\text{Velocity Ratio (VR)} = \frac{\text{distance moved by effort}}{\text{distance moved by load}} = \frac{2d}{d} = 2$$

Hence, VR = **2**.

19.

Ans: (a) it changes the direction of applied effort conveniently

Explanation:

A single fixed pulley does not multiply force (MA ≈ 1). It is mainly used so we can pull **downwards** to lift a load **upwards**, which is easier and more convenient.

Hence, option **(a)** is correct.

20.

Ans: (d) 0.5

Explanation:

Load L = mg = 10 × 10 = 100 N

Effort E = 200 N

Mechanical Advantage (MA)

$$\mathrm{MA} = \frac{L}{E} = \frac{100}{200} = 0.5$$

Thus, the mechanical advantage of the pulley is **0.5**.

21.

Ans: (c) Class I lever

Explanation:

In a **single fixed pulley**, the pulley's axle (fulcrum) is at the centre, with the load on one side of the rim and the effort on the other side—just like a **Class I lever** where the fulcrum lies between effort and load.

22.

Ans:

Given,

$$m = 400\,\mathrm{kg}, \quad g = 10\,\mathrm{m/s}^2$$
$$h = 19\,\mathrm{m}, \quad t = 40\,\mathrm{s}$$

Work done,

$$W = mgh = 400 \times 10 \times 19$$

Power of the pump,

$$P = \frac{W}{t} = \frac{400 \times 10 \times 19}{40}$$

$$P = 1900\,\mathrm{W}$$

Therefore, the power of the pump is 1900 W.